

When Is an Inverse a Function? - Practice

Algebra II Honors | Unit 1 | Lesson 12 Practice | TEK A2.2.C

Objective: I can restrict a quadratic with a vertex away from the origin and derive its inverse.

Section 1 — Recall

1. For $g(x) = (x - 2)^2$, find the vertex, and state a restriction that makes g one-to-one.

Vertex: _____ Restriction: _____

2. With restriction $x \geq 2$ on $g(x) = (x - 2)^2$, find $g^{-1}(x)$.

 $g^{-1}(x) =$ _____

Section 2 — Apply

3. For $h(x) = (x + 5)^2$, find the vertex, then find $h^{-1}(x)$ under the restriction $x \geq -5$.

Vertex: _____ $h^{-1}(x) =$ _____

4. For $k(x) = (x - 1)^2$ restricted to $x \leq 1$, find $k^{-1}(x)$.

 $k^{-1}(x) =$ _____

Section 3 — Challenge

5. Explain why the "2" in $g^{-1}(x) = 2 + \sqrt{x}$ (from $g(x) = (x - 2)^2$ restricted to $x \geq 2$) appears outside the radical, using the "undo in reverse order" idea.

6. For a general function $m(x) = (x - h)^2$ restricted to $x \geq h$, derive $m^{-1}(x)$ in terms of h .

 $m^{-1}(x) =$ _____

Section 4 — Explain It

7. A classmate argues that because both restrictions ($x \geq h$ and $x \leq h$) are "equally correct" for $m(x) = (x - h)^2$, the choice between them has absolutely no consequence - the two choices are functionally identical in every way. Is the classmate fully correct? Explain what actually does differ.
